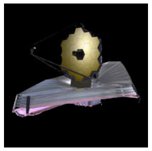

2352 - Wide Binary Stars in Nearby Dwarf Galaxies: A Novel Probe of Dark Matter on Subgalactic Scales

Cycle: 1, Proposal Category: AR

INVESTIGATORS

Name	*Institution*	*E-Mail*
Dr. Matthew G. Walker (PI)	**Carnegie Mellon University**	**mgwalker@andrew.cmu.edu**
Dr. Jorge Penarrubia (CoI) (ESA Member)	University of Edinburgh, Institute for Astronomy	jorpega@roe.ac.uk
Chris Kervick (CoI)	Carnegie Mellon University	ckervick@andrew.cmu.edu

OBSERVATIONS

ABSTRACT

The survival of widely separated binary stars within dense, low-mass dwarf galaxies depends strongly on the nature of dark matter. The standard Cold Dark Matter (CDM) model generically predicts that "ultrafaint" dwarf galaxy dark matter halos are composed of a smooth component with a centrally-divergent density "cusp", plus multitudes of self-bound subhalos, sub-subhalos, etc., down to a mass limit set by particle physics. Both smooth and clumpy components disrupt wide binary stars, via tidal forces and perturbative encounters, respectively, leaving an imprint on the binary separation function.

We propose to exploit JWST's unprecedented sensitivy to perform a search for wide binary stars (separation > 1000 A.U.) in the nearby ultrafaint dwarf galaxy Draco II, which offers favorable conditions for a search and will be observed as part of Early Release Science program 1334. We will apply Bayesian techniques to detect binary stars as closely separated pairs, and to infer the binary separation function directly from the distribution of stellar positions. We will interpret the results in the context of wide binary formation and survival in a dense dark matter halo environment, thereby providing novel tests of star formation in extreme environments and--especially if wide binaries are detected--a potentially definitive test of the CDM paradigm.

OBSERVING DESCRIPTION

This archival proposal will add a novel component to the legacy value of Eearly Release Science program 1334. It aims to use planned NIRCam observations of the ultrafaint Milky Way satellite Draco II in order to search for, and characterize the separations of, wide binary stars in that system. The wide binary separation function is sensitive to the abundance of subgalactic dark matter halos that are a generic prediction of the Cold Dark Matter paradigm, and also to the structure of the smooth component of Draco II's dark matter halo. Thus the inferred wide binary separation function will be used to constrain the distribution of dark matter on subgalactic scales within Dra II, providing a novel test of the cold dark matter model.